Fiscal Year:	FY 2022	**Task Last Updated:**	FY 09/29/2021
PI Name:	Weber, Richard Ph.D.		
Project Title:	Microgravity Investigation of Thermophysical Properties of Supercooled Molten Metal Oxides		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Arlington Heights	**State:**	IL
Zip Code:	60004-7708	**Congressional District:**	8
Comments:	NOTE: Formerly with Containerless Research, Inc., until 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	05/21/2017	**End Date:**	05/20/2022
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:	ISS		
Flight Assignment:	NOTE: New grant number 80NSSC18K0059 awarded with period of performance of 11/9/2017-5/20/2022; added to original grant since is continuation of the work (Ed., 5/18/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	80NSSC18K0059 ; NNX17AH73G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research is implementing microgravity based experiments using the JAXA (Japanese Aerospace Exploration Agency) flight Electrostatic Levitation Furnace (ELF) to achieve precise measurements of the thermophysical properties of supercooled (also called undercooled) molten metal oxides. The work will include both ground based and flight experiments on liquids formed from aluminum, calcium, silicon, and rare earth element oxide mixtures. These liquids are precursors to high value-added glass materials that are used in photonics, lasers, optical communications, and imaging applications. The outcome of the proposed international research collaboration will be to: (i) Measure technologically important thermophysical properties of normal and supercooled liquid metal oxides of interest in connection with production of photonic and optical device materials, (ii) Develop thermophysical property datasets that complement existing atomic structure data on the materials under investigation. Use the resulting data to increase understanding of the nature of fragile liquids and the glass transition, and (iii) Disseminate the new data via publication, through NASA's Physical Sciences Informatics System and the Materials Genome Initiative to help accelerate the development of new products. We will investigate the behavior of liquid phase metal oxides as they transform to glass in quiescent, diffusion-controlled (i.e., without fluid motion) conditions. This work is needed to optimize photonic and optical device materials to obtain uniform properties and stable performance in applications. Overall the work will advance the broader goals of optimizing computational modeling of materials processes and developing superior materials by design. The flight ELF can achieve pristine molten and supercooled samples. Such samples can be studied using non-contact measurements of temperature, volume, surface tension, and viscosity. Spherical samples ca. 2 mm in diameter suitable for experiments in the flight ELF will be made by laser hearth melting. Prior ground-based work demonstrated that the sample materials can be: (i) levitated using electrostatic methods, (ii) heated using laser beams, and (iii) melted without significant evaporation. Proposed ground-based work will push the limits of these measurements and help to optimize the methods for analysis for the flight data. ED. NOTE 5/18/2020: New grant number 80NSSC18K0059 awarded with period of performance of 11/9/2017-5/20/2022; added to original grant since is continuation of the work.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work relates to development, characterization, and optimization of functional glasses and glass ceramics made in highly non-equilibrium conditions. These materials are high value add products that are using in telecommunications, displays (TVs and phones), medical lasers, and security sensor equipment. The project work includes outreach to undergraduate and high school students.

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Reporting period: 11/20/2020-11/19/2021

This project is investigating the thermophysical properties of molten metal oxides. These materials are of critical importance in a wide range of applications including functional glasses, batteries, next generation propulsion systems, lasers and optical products. An increased understanding of oxide materials is of fundamental value in materials science and applied value in developing means to harness planetary materials for use in space exploration. Fundamental research provides an essential underpinning for transformational discovery and development of major new applications. This project aims to measure the viscosity, density and surface tension of a variety of molten oxides, some of which form glass. Containerless conditions are essential in order to achieve deep supercooling without heterogeneously induced nucleation of crystals. This requirement will be met by using the Japanese Aerospace Exploration Agency (JAXA) Electrostatic Levitation Furnace (ELF). The ELF instrument is an electrostatic levitator designed for work on metal oxides at temperatures up to about 2500ºC in containerless conditions. By working in microgravity it is expected that nearly quiescent molten samples can be obtained. Induced oscillations of levitated liquid droplets is used to study their fluid properties. Image analysis reveals the oscillation amplitude and power spectrum, from which the required properties (surface tension and viscosity) are calculated. The materials of primary interest are molten metal oxides, many of which form fragile liquids that can be made into glass provided the melts can be deeply supercooled. In some cases there is already considerable data on the atomic structure of these melts. An important goal of this research is to correlate the structure and properties (in particular viscosity and density) to advance models of the processing of materials. These models are important in advancing basic science of liquids and glasses and in development of high value-added applied materials.

1. Status: The project team includes Materials Development Inc. (MDI) scientists and Professors Robert Hyers, University of Massachusetts, Amherst, Jonghyun Lee, Iowa State Univeristy, Dr. Shinji Kohara, National Institute for Materials Science, Japan, and Dr. Takehiko Ishikawa, Japanese Aerospace Exploration Agency, Japan. The collaborators from Japan are not funded from this grant.

The grant was awarded on 5/21/2017 and assigned a new contract number in November 2017. The period of performance was extended from four to five years and the budget was reduced. First year funding was at a level sufficient to enable scoping of the work and preliminary ground-based experiments in preparation for a more detailed experimental campaign during the subsequent years. Second year funding was also at a reduced level sufficient for planning and starting to prepare for the Science Concept Review (SCR). The SCR was successfully completed in March 2020 and remaining issues from the SCR were addressed.

2. Technical Activities: Activities during the third year included: (i) synthesis of final set of test samples that were sent to JAXA for ground-based testing prior to final selection, (ii) launching samples to the International Space Station (ISS), (iii) ground-based structure measurements on glass samples, and (iv) planning for flight experiments.

(i) The candidate compositions approved during the SCR. Two gram batches of each composition were made from high purity metal oxide powders. A set of 2 mm diameter test samples were sent to JAXA on 7/7/2020 for ground-based testing. On 8/17/2020, JAXA confirmed that all compositions passed the vibration test. Additional ground-based tests were performed, including laser beam heating at 980 nm and checking effects of humidity on sample stability. All candidate compositions except LiBO2 passed the humidity test. For compositions that did not contain optically active ions (CA, CS, MS, and mullite), 2 wt% Yb2O3 was added to increase laser beam absorption at the 980 nm wavelength.

Initially the plan was to fly three each of 15 compositions for a total of 45 samples. Due to some early difficulties in the levitation experiments, JAXA requested that MDI instead launch five each of nine compositions. The final samples were launched in June 2021.

Task Progress:	(ii) A set of 45 samples comprising five of each composition was launched to ISS onboard SpX-22 on June 3, 2021. Samples were located in three different holders, each with 15 selected compositions. Experiments are planned to begin in December 2021. (iii) Ground-based tests are in progress to characterize the final samples. In addition to the flight samples, extra 2 mm diameter beads were synthesized at MDI using aerodynamic levitation and laser beam heating. These samples were mounted, polished, and analyzed with scanning electron microscopy and energy dispersive X-ray spectroscopy, which showed that all samples were within measurement error of their nominal composition. For those compositions that form glass, atomic structure was measured using high energy X-ray diffraction at the Advanced Photon Source, Sector 6-ID-D. Neutron diffraction measurements have also been completed using the NOMAD instrument at the Spallation Neutron Source. X-ray and neutron diffraction provide complementary probes of atomic structure, since X-rays are more sensitive to heavier elements (i.e., cations) and neutrons are sensitive to scattering from oxygen. Diffraction data have been collected for all ground-based glasses, and data analysis is ongoing. The atomic structure data will be used in interpretation of the results of the flight experiments, which will enable a more thorough understanding of how glass-forming liquids behave when they are supercooled. As one example of relating structural measurements with liquid phase processing, we used a combined small- and wide-angle X-ray scattering technique to study correlations between liquid-liquid phase separation and atomic coordination in aluminosilicates, since mullite (Al6Si2O13) is one of the flight samples. Phase separation is desirable for some technologies (e.g., glass ceramics) but not for others (e.g., low-loss fiber optics), so understanding the relationship between material processing and structure will help advance these technologies. Aluminosilicate glasses (10-50 mol. % Al2O3) were prepared from high-temperature melts in containerless conditions, spanning the range of L-L immiscibility. In addition to structural studies, work is ongoing to evaluate the critical cooling rates for glass-forming samples and to validate the temperature corrections for optical pyrometry, since at lower temperature some compositions may become partially transparent at the pyrometer wavelength. Glass samples are being prepared for analysis using differential scanning calorimetry and solution calorimetry at the Materials of the Universe center at Arizona State University under an ongoing collaboration. These thermal analyses will provide information about samples' glass transition temperatures, crystallization behavior, and enthalpy of vitrification. Lastly, nuclear magnetic resonance (NMR) measurements are being planned in collaboration with Corning Incorporated, which will provide additional information on local structure for elements that can be probed with NMR (e.g., Si and Al). (iv) In planning for the flight experiments, the expected oscillation frequencies of molten samples were estimated using literature data for surface tension. Data were typically not available for the exact composition under study, so the values were obtained by extrapolating or interpolating data for similar materials from the literature [1-6]. The estimated oscillation frequencies provide a useful starting point for measurements in ELF. The initial coarse scan of frequencies used to obtain the resonant frequency of a liquid drop will be based on bracketing the estimated surface tension by +/- 100 mN/m. The typical frequency range is from about 120-160 Hz. MDI has begun development of image analysis procedures and programming scripts (Python) for reduction of the temperature and video data anticipated from ELF measurements. Using example data files provided by JAXA, we have identified the correct software programs (Fiji, Adobe Premier Pro, LabView) and steps required to streamline data processing and analysis. MDI plans to send two staff members to JAXA's control center in Tsukuba for a period of one week during the experiments. While work can be conducted effectively in remote mode, a short site visit will help to enhance collaboration and allow more detailed discussions about the work. Travel will be planned based on current restrictions that might result from COVID-19. References: [1] Kingery, J. Am. Ceram. Soc., 42, 6, (1959). [2] Bates, et al., Contract report NAS 3-14308, (1972). [3] Choi, et al., ISIJ International, 42, 221 (2002). [4] Weirauch, et al., J. Am. Ceram. Soc., 79, 920 (1996). [5] Conte, et al., Materials, 11, 2475 (2018). [6] Bates, et al., Contract report NAS 3-14308, (1972). A logo for the project was designed in collaboration with NASA. This is being produced as stickers and sew-on patches. A wall hanging version is being displayed at the JAXA ELF control room along with logos for other ELF experiments.
Bibliography Type:	Description: (Last Updated: 06/06/2026)
Articles in Peer-reviewed Journals	Benmore C, Gonzalez G, Alderman O, Wilke S, Yarger J, Leinenweber K, Weber R. "Hard x-ray methods for studying the structure of amorphous thin films and bulk glassy oxides." J Phys Condens Matter. 2021 Apr 23;33(19):194001. https://doi.org/10.1088/1361-648X/abe352 ; PMID: 33540391 , Apr-2021